Online Purchasing Intention Analysis

FINAL REPORT

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Link For Presentation- https://cometmail-my.sharepoint.com/:v:/g/personal/pxh220007_utdallas_edu/EUKYNYGgLqpKgqatJfa5r5kBwMnlUsZz0pweLzE45fVBBg?e=vWuA3T

1. Executive Summary

The objective of this analysis is to provide a reliable and feasible recommendation algorithm to predict shopper behaviour. Using pageview data recorded during the visit, certain session and user data, and the visitor's predicted purchasing intent. The plan is to use clustering and maybe classification techniques to be able to make predictions about shoppers' intentions. Together, our models are used to identify users who are interested in making a purchase but are likely to depart the site within the prediction horizon and to take appropriate action to reduce website abandonment and increase conversion rates for online purchases. Our results demonstrate the viability of precise and scalable clickstream and session information data-based purchasing intention prediction for virtual retail environments.

1.1. Project Motivation/Background

To analyse and classify customers based on their intention to complete the purchase using customer attributes and their online behaviour. To find out the seasonality trends of customers visiting the website/purchasing products relating to festivals, occasions, weekends, and times of the year and identifying potential customers based on customer profiles for marketing & promotional campaigns. Targeting new customers based on a typical profile identified by current visiting customers.

1.2. Data Description

We have used second-hand data from the UCI repository. The dataset comprises customers' information during their distinct sessions on an E-commerce website. It consists of attribute vectors from 12,330 sessions recorded over one year to avoid any inclination toward a particular campaign, holiday, user profile, or timing. The dataset consists of 10 numerical and 8 categorical attributes.

Numerical Attributes	Attribute Description
Administrative	Number of account management pages that the visitor visited
Administrative_Duration	The total time (in seconds) the visitor spent on pages related to account management
Informational	The number of pages visited by the visitor about the Web site, communication of the site.
Informational_Duration	The total amount of time the visitor spent on informational pages (in seconds)
ProductRelated	Number of product-related pages that visitors have visited
ProductRelated_Duration	The total amount of time (in seconds) the visitor spent on the products page
BounceRates	Percentage of web users that enter the site from that page and then leave
ExitRates	Percentage of pageviews that were the last in the session
PageValues	The typical value for a web page a user visits before making an online purchase
SpecialDay	The proximity of the site visitation period to a particular holiday

Categorical Attributes	Attribute Description
OperatingSystems	OS information of the visitor
Browser	Browser information of the visitor
Region	Geolocation of the visitor
TrafficType	The website's traffic source via which the visitor arrived (e.g., SMS, direct)
VisitorType	Type of visitor (e.g., New Visitor, Returning Visitor, and Other)
Weekend	Boolean value indicating if it's a weekend
Month	Month value
Revenue	Class label showing if a transaction has been completed following the visit

2. Exploratory Data Analysis

2.1. Preliminary Data Exploration

The Preliminary analysis of entire dataset is necessary to get a basic overview of the variable and their statistical information.

DIM (), HEAD (), STR (), and SUMMARY () provides us quick data structure previews.

```
> dim(shoppers.df)
[1] 12330    18
> head(shoppers.df)
  Administrative Administrative_Duration Informational Informational_Duration
1              0                       0             0                      0
2              0                       0             0                      0
3              0                       0             0                      0
4              0                       0             0                      0
5              0                       0             0                      0
6              0                       0             0                      0
  ProductRelated ProductRelated_Duration BounceRates ExitRates PageValues
1              1                0.000000  0.20000000 0.2000000          0
2              2               64.000000  0.00000000 0.1000000          0
3              1                0.000000  0.20000000 0.2000000          0
4              2                2.666667  0.05000000 0.1400000          0
5             10              627.500000  0.02000000 0.0500000          0
6             19              154.216667  0.01578947 0.0245614          0
  SpecialDay Month OperatingSystems Browser Region TrafficType
1          0   Feb                1       1      1           1
2          0   Feb                2       2      1           2
3          0   Feb                4       1      9           3
4          0   Feb                3       2      2           4
5          0   Feb                3       3      1           4
6          0   Feb                2       2      1           3
        VisitorType Weekend Revenue
1 Returning_Visitor   FALSE   FALSE
2 Returning_Visitor   FALSE   FALSE
3 Returning_Visitor   FALSE   FALSE
4 Returning_Visitor   FALSE   FALSE
5 Returning_Visitor    TRUE   FALSE
6 Returning_Visitor   FALSE   FALSE
```

```
> str(shoppers.df)
'data.frame':	12330 obs. of  18 variables:
 $ Administrative         : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Administrative_Duration: num  0 0 0 0 0 0 0 0 0 0 ...
 $ Informational          : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Informational_Duration : num  0 0 0 0 0 0 0 0 0 0 ...
 $ ProductRelated         : int  1 2 1 2 10 19 1 0 2 3 ...
 $ ProductRelated_Duration: num  0 64 0 2.67 627.5 ...
 $ BounceRates            : num  0.2 0 0.2 0.05 0.02 ...
 $ ExitRates              : num  0.2 0.1 0.2 0.14 0.05 ...
 $ PageValues             : num  0 0 0 0 0 0 0 0 0 0 ...
 $ SpecialDay             : num  0 0 0 0 0 0 0.4 0 0.8 0.4 ...
 $ Month                  : chr  "Feb" "Feb" "Feb" "Feb" ...
 $ OperatingSystems       : int  1 2 4 3 3 2 2 1 2 2 ...
 $ Browser                : int  1 2 1 2 3 2 4 2 2 4 ...
 $ Region                 : int  1 1 9 2 1 1 3 1 2 1 ...
 $ TrafficType            : int  1 2 3 4 4 3 3 5 3 2 ...
 $ VisitorType            : chr  "Returning_Visitor" "Returning_Visitor" "Returning_Visitor" "Returning_Visitor" ...
 $ Weekend                : logi  FALSE FALSE FALSE FALSE TRUE FALSE ...
 $ Revenue                : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
```

```
> summary(shoppers.df)
 Administrative   Administrative_Duration Informational
 Min.   : 0.000   Min.   :   0.00         Min.   : 0.0000
 1st Qu.: 0.000   1st Qu.:   0.00         1st Qu.: 0.0000
 Median : 1.000   Median :   7.50         Median : 0.0000
 Mean   : 2.315   Mean   :  80.82         Mean   : 0.5036
 3rd Qu.: 4.000   3rd Qu.:  93.26         3rd Qu.: 0.0000
 Max.   :27.000   Max.   :3398.75         Max.   :24.0000
 Informational_Duration ProductRelated    ProductRelated_Duration
 Min.   :   0.00        Min.   :  0.00    Min.   :    0.0
 1st Qu.:   0.00        1st Qu.:  7.00    1st Qu.:  184.1
 Median :   0.00        Median : 18.00    Median :  598.9
 Mean   :  34.47        Mean   : 31.73    Mean   : 1194.8
 3rd Qu.:   0.00        3rd Qu.: 38.00    3rd Qu.: 1464.2
 Max.   :2549.38        Max.   :705.00    Max.   :63973.5
  BounceRates          ExitRates          PageValues          SpecialDay
 Min.   :0.000000   Min.   :0.00000   Min.   :  0.000   Min.   :0.00000
 1st Qu.:0.000000   1st Qu.:0.01429   1st Qu.:  0.000   1st Qu.:0.00000
 Median :0.003112   Median :0.02516   Median :  0.000   Median :0.00000
 Mean   :0.022191   Mean   :0.04307   Mean   :  5.889   Mean   :0.06143
 3rd Qu.:0.016813   3rd Qu.:0.05000   3rd Qu.:  0.000   3rd Qu.:0.00000
 Max.   :0.200000   Max.   :0.20000   Max.   :361.764   Max.   :1.00000
    Month           OperatingSystems     Browser           Region
 Length:12330       Min.   :1.000    Min.   : 1.000   Min.   :1.000
 Class :character   1st Qu.:2.000    1st Qu.: 2.000   1st Qu.:1.000
 Mode  :character   Median :2.000    Median : 2.000   Median :3.000
                    Mean   :2.124    Mean   : 2.357   Mean   :3.147
                    3rd Qu.:3.000    3rd Qu.: 2.000   3rd Qu.:4.000
                    Max.   :8.000    Max.   :13.000   Max.   :9.000
  TrafficType    VisitorType          Weekend          Revenue
 Min.   : 1.00   Length:12330       Mode :logical   Mode :logical
 1st Qu.: 2.00   Class :character   FALSE:9462      FALSE:10422
 Median : 2.00   Mode  :character   TRUE :2868      TRUE :1908
 Mean   : 4.07
 3rd Qu.: 4.00
```

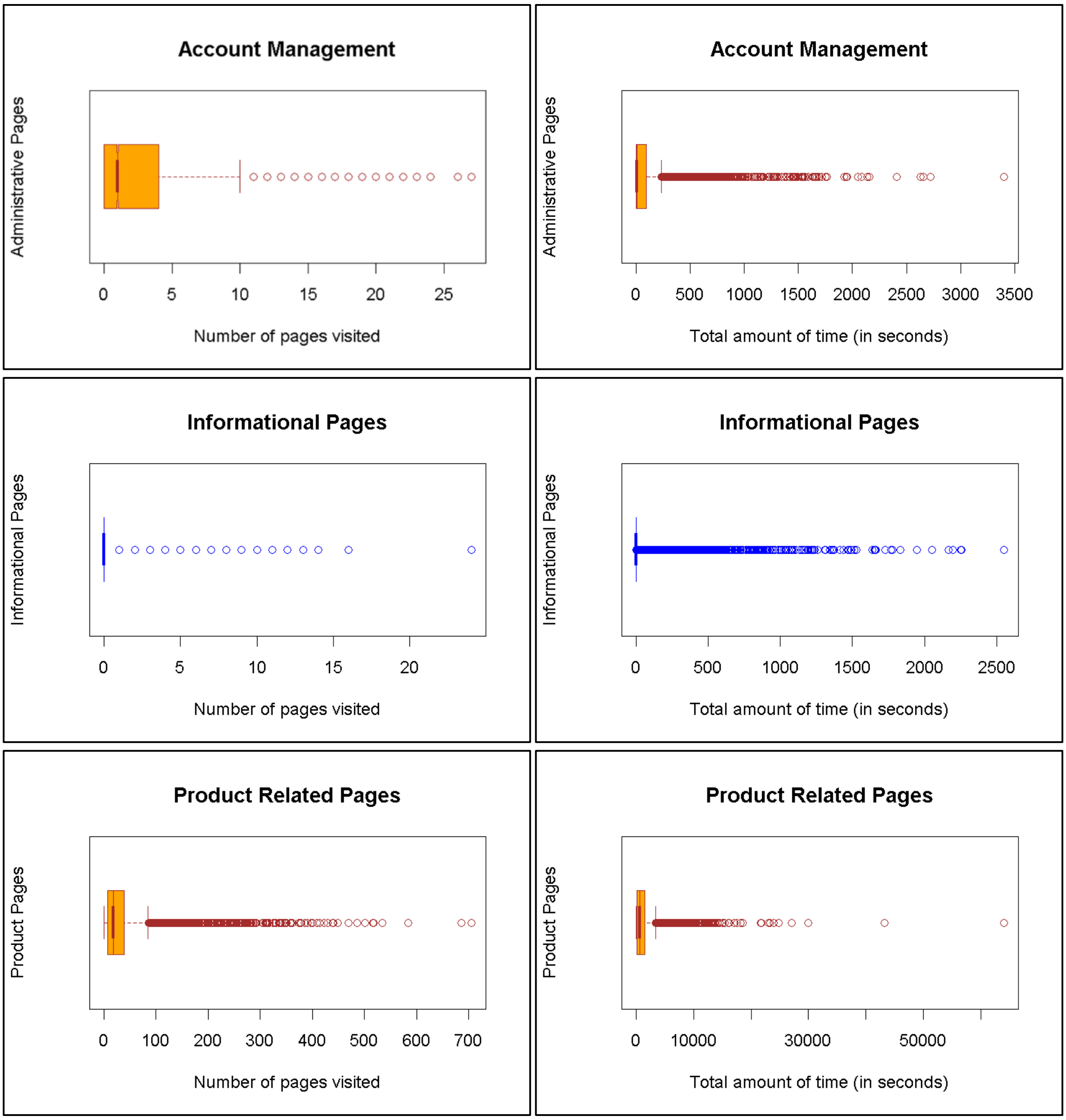
Finding out about missing values:

```
> colSums(is.na(shoppers.df))
         Administrative Administrative_Duration           Informational
                      0                       0                       0
 Informational_Duration          ProductRelated ProductRelated_Duration
                      0                       0                       0
            BounceRates               ExitRates              PageValues
                      0                       0                       0
             SpecialDay                   Month        OperatingSystems
                      0                       0                       0
                Browser                  Region             TrafficType
                      0                       0                       0
            VisitorType                 Weekend                 Revenue
                      0                       0                       0
> sum(is.na(shoppers.df))
[1] 0
```

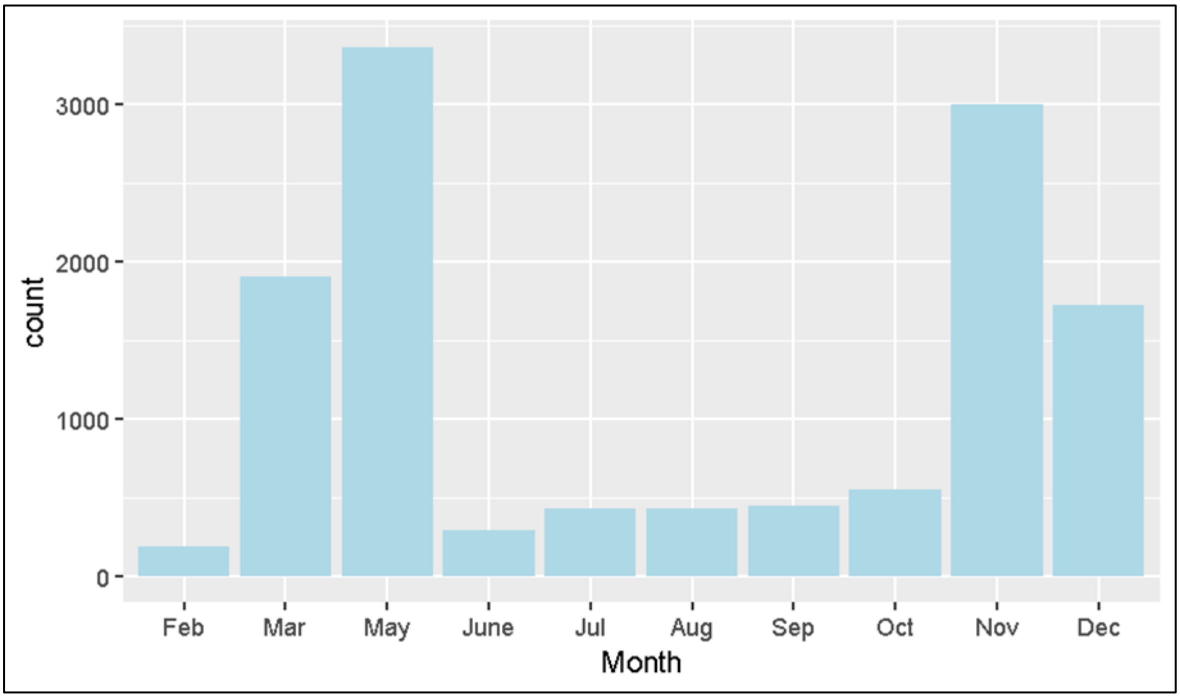
We can observe that there are no missing figures because this information is based on website visits. Each data point is the result of a user interacting with the website in some way; if there are no interactions, the data point is a "0".

2.2. Exploratory Data Analysis

Plotting Box Plots of the different pages visited by the user and the time spent on those pages.

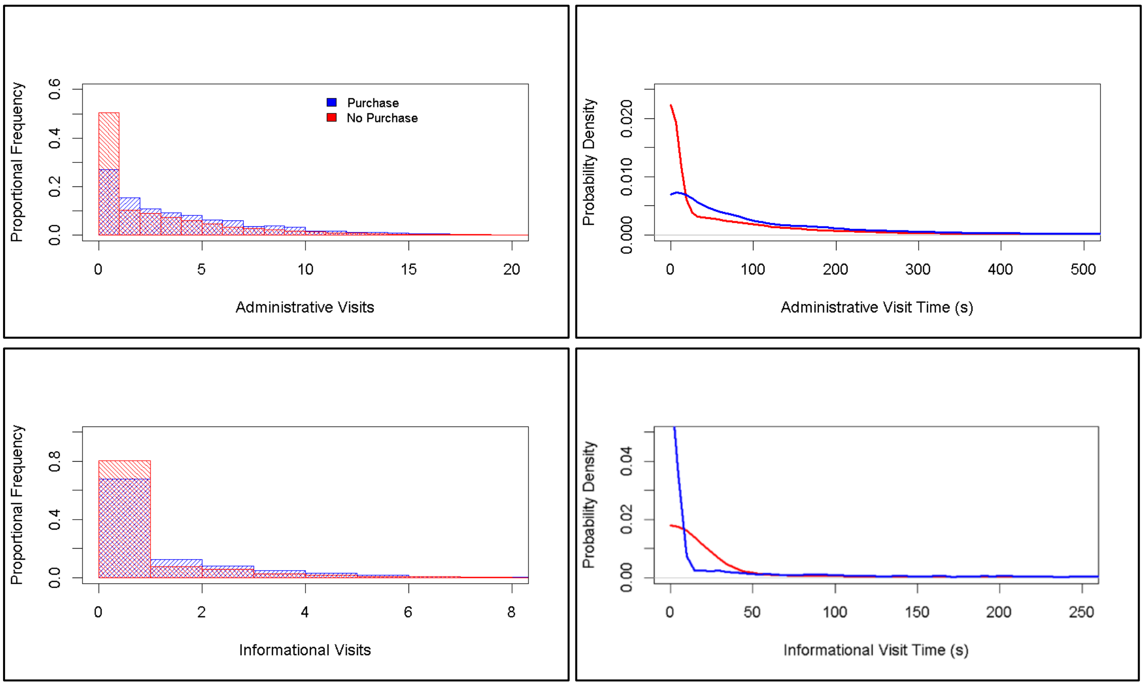


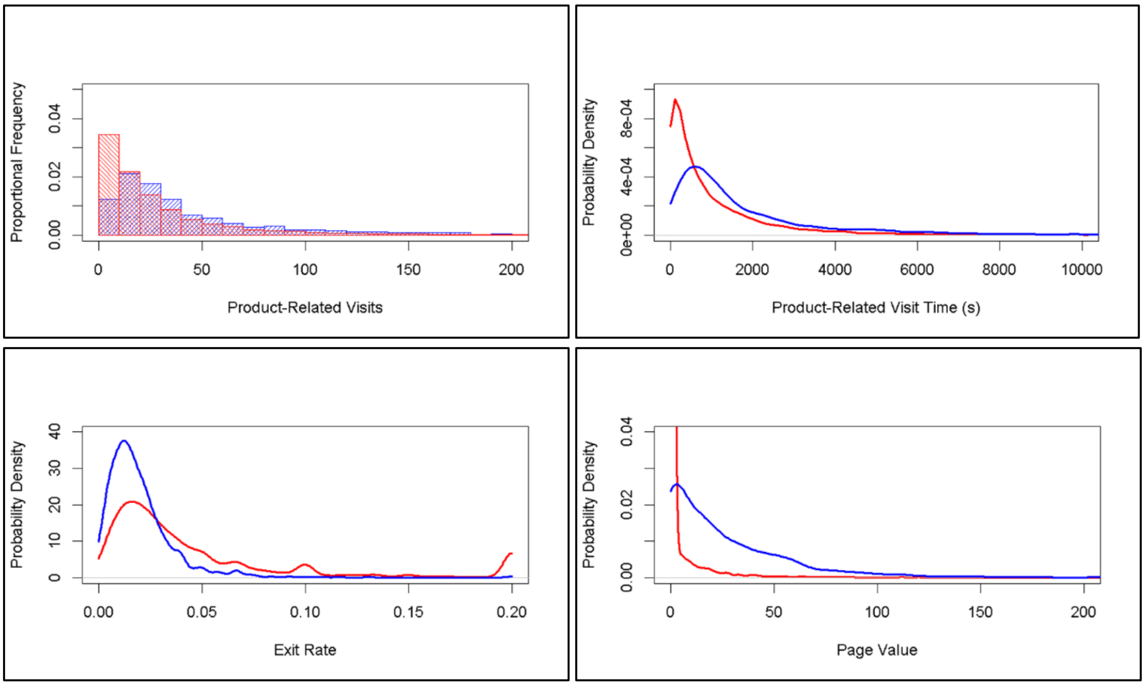
These box plots show that there aren't any significant outliers in the data.



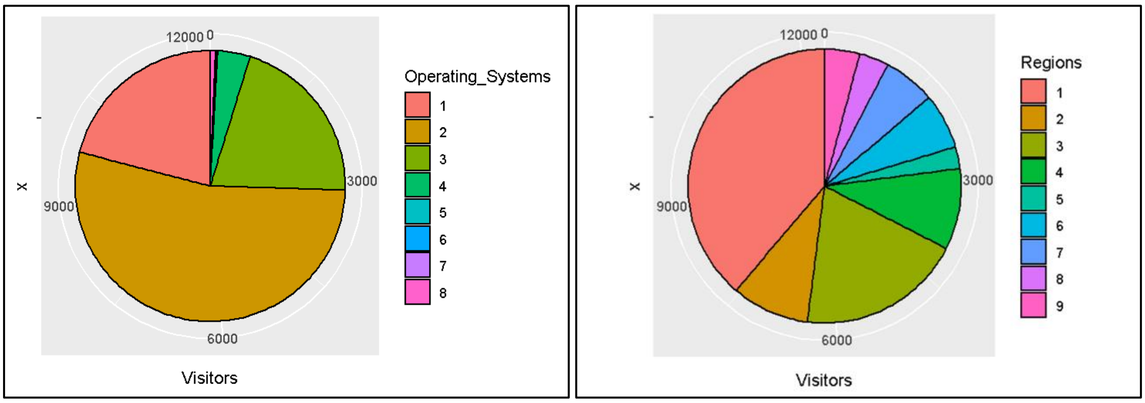
In September, October, and November, which are traditionally associated with America's "shopping season," we observe extremely high shopping rates. Notable is also the high number of website visitors in May.

We examine the tracking data visually to determine whether there are any distinctions between shoppers and non-shoppers. We do this by plotting the tracking data against revenue generated.





Plotting pie charts for operating systems and regions of the users.



According to the pie chart above, type 2 OS is used by most website visitors followed by type-3 and type-1 respectively.

Also, we can infer from the pie chart-2 that many visitors come from area 1.

3. Data Mining Tasks/Models

We are going to perform several data mining models such as K-means clustering, Hierarchical Clustering, Classification, Logistic Regression and Neural Networks to our data set to help us find insights about purchasing intentions of the users.

3.1. K-Means Clustering

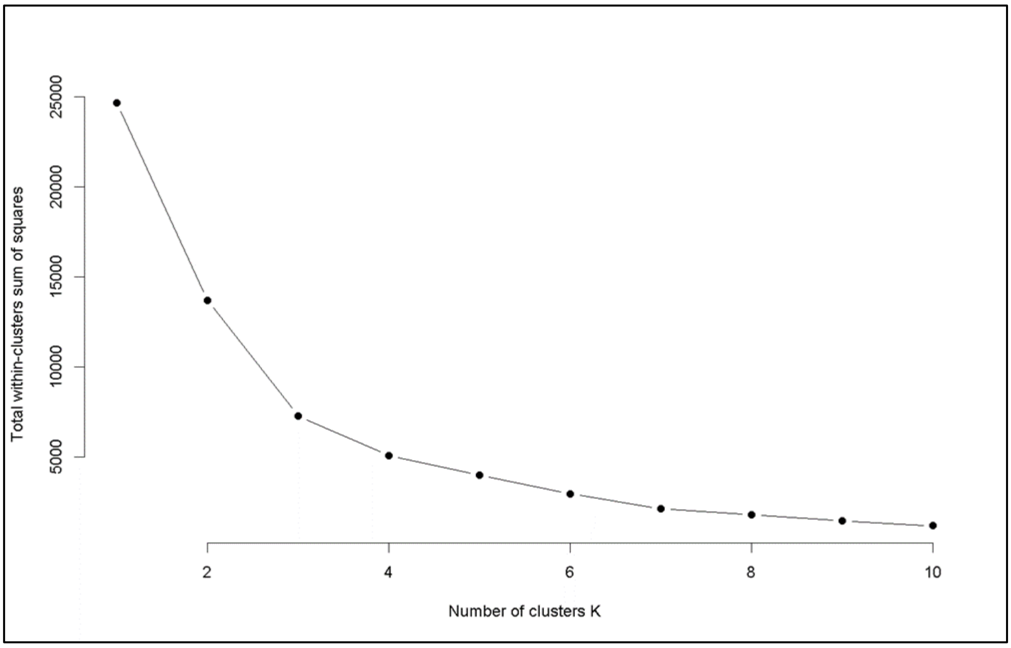
Our data visualization indicates that there are no obvious patterns in the distribution of our variables, and as a result, clustering may be a suitable sorting algorithm for our requirements. It will examine the information and hunt for clusters.

We are using ProductRelated Duration and BounceRates to find targeted customers.

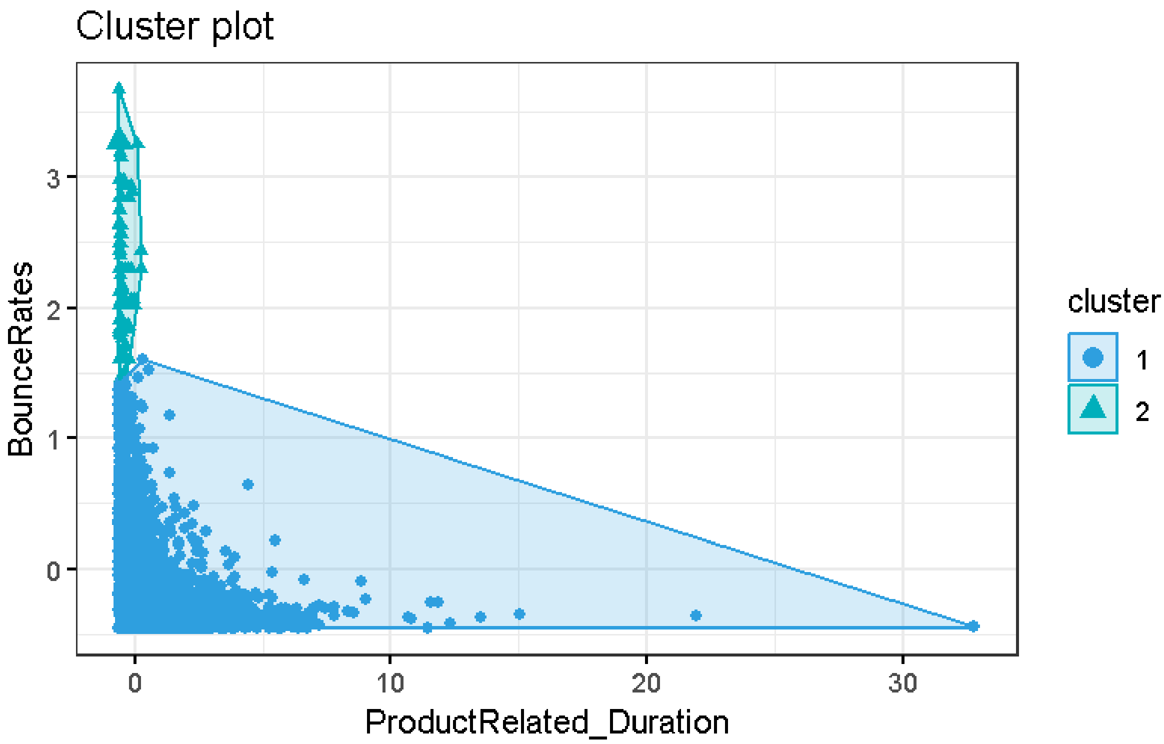
3.1.1. Diagrams

K-elbow Method

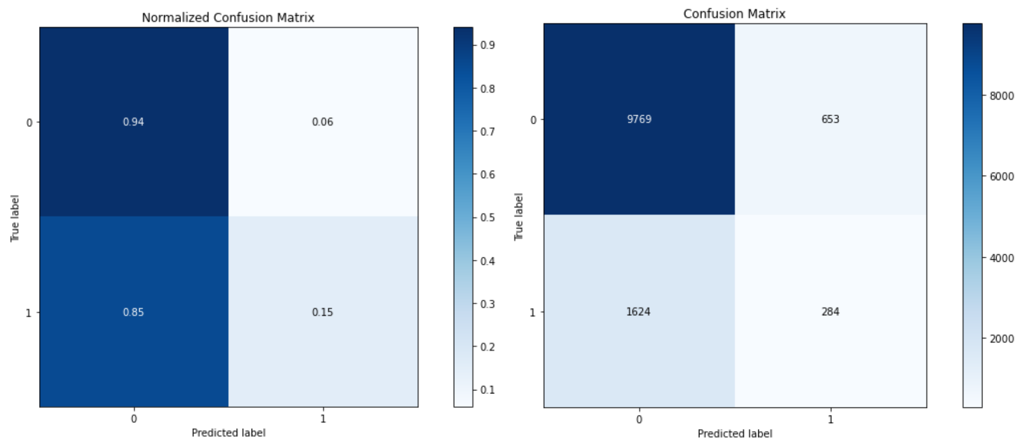
The maximum bend is at the second index, which means that there are two groups that may be optimally clustered for the variables ProductRelated Duration and BounceRates. We use the KMeans approach after determining the clustering number, then depict the clusters.



Visualizing our K-means Clusters



We plot the confusion matrix against the revenue



3.1.2. Findings

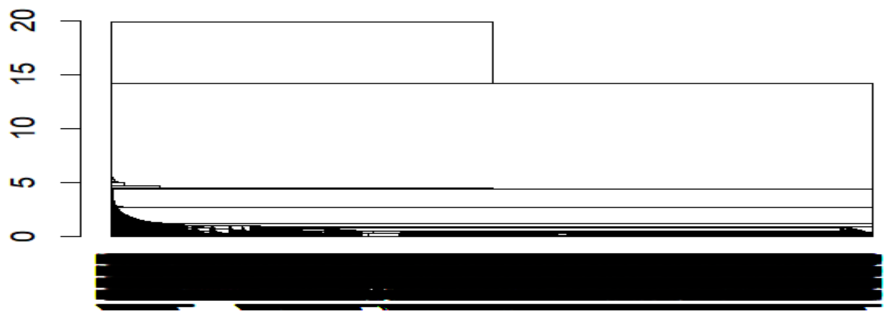
The confusion matrix reveals that 9,769 of the 10,422 missed revenues, or 94% of them, are clustered as disinterested consumers. Only 284, or 15%, of the 937 successful revenues are grouped as target clients. Additionally, the adjusted index score isn't particularly high. It is obvious that we misclassified a lot of profitable revenue-generating sessions as disinterested consumers, which suggests that even with a high bounce rate and a short time spent.

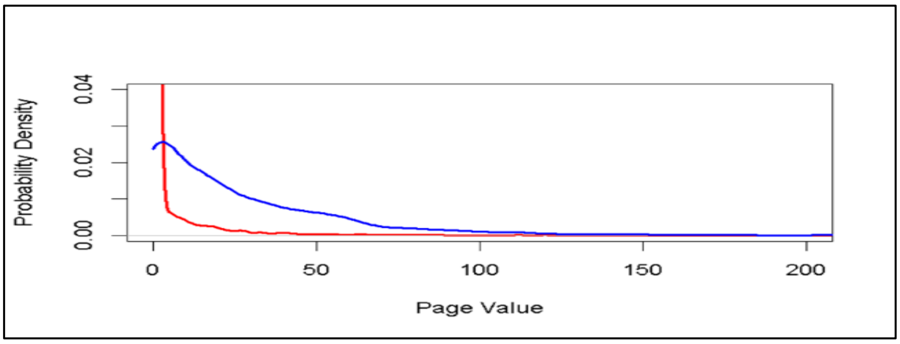
3.2. Hierarchical Clustering

Hierarchical clustering is an agglomerative approach which starts with n clusters and combines the similar records to form similar clusters. We used the average linkage for computing the distance between the clusters.

For clustering, we manipulated the categorical variables and then computed the distance matrix. The benefit of hierarchical clustering is that we can change the number of clusters or the detail/ depth of the distinguishing factors by cutting the tree at desired level.

3.2.1. Diagrams





3.2.2. Findings

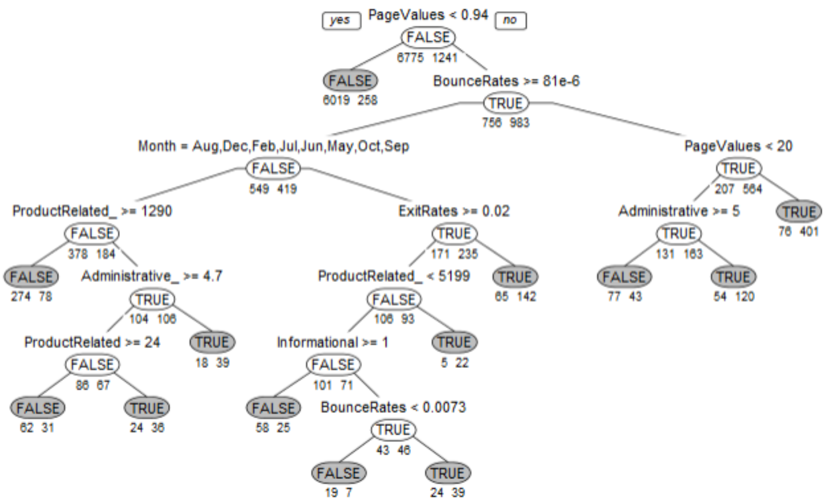
The clusters formed had high number of records within each cluster. The top two clusters after observing the values, had similar revenue generated and similar page values. So, the clustering divided the revenue- generating cluster into sub clusters with different properties which can be helpful in tracking a particular group of customers who could be potential revenue generating customers.

3.3. Classification

We have Performed the Decision Tree Classification analysis on our dataset. We have worked on building a model which predicts if the transaction has been completed following the visit by a user on the Online Shopping website. We Considered the (REVENUE) as the Target variable which gives output in Boolean Values (True/False), if True, then the purchase has been completed. We Performed the cross validation, wherein we found that the optimal complexity

Parameter is 0.055 with Number of splits= 13. We were able to achieve the accuracy of 90.8% after the Pruning process. We observed that the (PAGEVALUE) is a variable which was used to perform the root node splitting.

3.3.1. Diagrams



3.3.2. Findings

From the above results, we can interpret that as a web developer of the online Store website, one should focus on increasing the page value, decreasing the bounce rate, and paying attention to the types of products listed on pages of administrative type ‘5’ and below.

3.4. Logistics Regression

We have performed the Logit Regression for our Online Retail dataset. We included all 18 input variables for this data mining process. As we look to predict if the Revenue was generated after a user session. Revenue is the Target Variable and others are the input variables. If there is a transaction, then the output is **TRUE** else **FALSE**. For this process we split the entire dataset into Train and validation dataset, where the split ratio is 0.65 for Train and 0.35 for test. We Run the Regression by implementing the glm () function. With the summary () function we print the coefficients of regression and their corresponding p-values. Lower the P-value more significant is the variable. We implement the prediction function to predict if the Revenue has been generated using the validation dataset. As the predicted out-put is Numeric, we can interpret that using an ifelse statement, such that if the predicted value is >0.5 then **TRUE** else **FALSE**. We draw a confusion matrix as an intuitive metrics to find the accuracy of the model. In this case the Categories being **TRUE** and **FALSE**

3.4.1. Diagrams

```
> summary(backwards)

Call:
glm(formula = Revenue ~ ProductRelated_Duration + ExitRates + 
    PageValues + Month + Browser + VisitorType, family = "binomial", 
    data = trainset)

Deviance Residuals: 
    Min       1Q   Median       3Q      Max  
-6.1773  -0.4766  -0.3314  -0.1624   3.2503  

Coefficients:
                              Estimate  Std. Error z value             Pr(>|z|)    
(Intercept)                -1.67000113  0.21501656  -7.767  0.00000000000000805 ***
ProductRelated_Duration     0.00011451  0.00001674   6.842  0.00000000000783009 ***
ExitRates                 -17.54980714  2.00966415  -8.733 < 0.0000000000000002 ***
PageValues                  0.08198768  0.00296024  27.696 < 0.0000000000000002 ***
MonthDec                   -0.72817548  0.22121755  -3.292             0.000996 ***
MonthFeb                   -2.01179406  0.79333427  -2.536             0.011217 *  
MonthJul                    0.16658568  0.25998101   0.641             0.521678    
MonthJune                  -0.61056127  0.35350728  -1.727             0.084140 .  
MonthMar                   -0.68634787  0.22019616  -3.117             0.001827 ** 
MonthMay                   -0.67913783  0.20497971  -3.313             0.000922 ***
MonthNov                    0.45462915  0.19597938   2.320             0.020353 *  
MonthOct                   -0.00911762  0.24213710  -0.038             0.969963    
MonthSep                    0.18330374  0.25319972   0.724             0.469097    
Browser                     0.04172301  0.02348084   1.777             0.075585 .  
VisitorTypeOther           -0.75669964  0.66522251  -1.138             0.255324    
VisitorTypeReturning_Visitor -0.29670030  0.10656886  -2.784             0.005367 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6909.3  on 8015  degrees of freedom
Residual deviance: 4690.9  on 8000  degrees of freedom
AIC: 4722.9

> backwards.reg.pred <- predict(backwards, validationset, type = "response")
> backwards.reg.pred.classes <- ifelse(backwards.reg.pred > 0.5, TRUE, FALSE)
> confusionMatrix(as.factor(backwards.reg.pred.classes), as.factor(validationset$Revenue))
Confusion Matrix and Statistics

          Reference
Prediction FALSE TRUE
     FALSE  3553  405
     TRUE     94  262
                                               
               Accuracy : 0.8843               
                 95% CI : (0.8744, 0.8937)     
    No Information Rate : 0.8454               
    P-Value [Acc > NIR] : 0.0000000000001223   
                                               
                  Kappa : 0.4534               
                                               
 Mcnemar's Test P-Value : < 0.00000000000000022
                                               
            Sensitivity : 0.9742               
            Specificity : 0.3928               
         Pos Pred Value : 0.8977               
         Neg Pred Value : 0.7360               
             Prevalence : 0.8454               
         Detection Rate : 0.8236               
   Detection Prevalence : 0.9175               
      Balanced Accuracy : 0.6835               
                                               
       'Positive' Class : FALSE                
```

3.4.2. Findings

formula = Revenue ~ ProductRelated_Duration + ExitRates +

PageValues + Month + Browser + VisitorType

On performing the Backward stepwise variable selection, we can conclude that ProductRelated_Duration, ExitRates, PageValue, Month, Browser and Visitor Type are some of the Significant input variable which needs to be considered.

3.5. Neural Networks

The next supervised learning approach we explored was neural networks. Neural networks work like a black box wherein the inputs are weighted and produce a predicted output value based on the adjusted weight values for better accuracy.

In our dataset, we found that the most accurate results are obtained by certain inputs that have a significant impact in predicting the results.

Revenue has a binary value i.e., The customer either makes a purchase or doesn't. The input parameters PageValues, BounceRates, ExitRates and Weekend produce a predicted result for Revenue. After splitting the dataset into training and testing data, we used the training dataset to train the neural network and used the testing data to validate the predicted results.

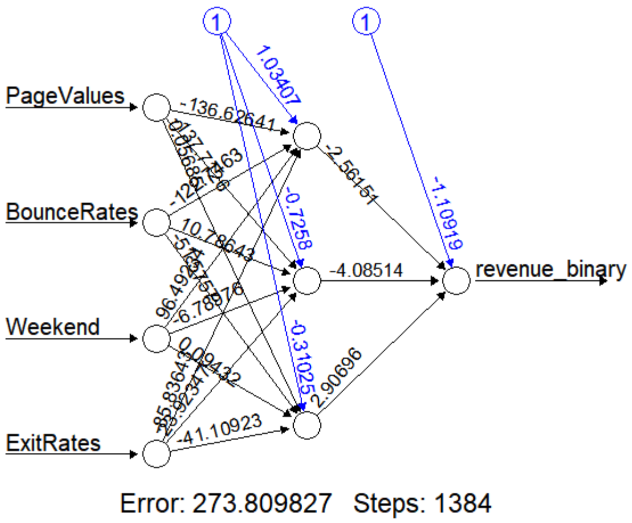
R-CODE:

nn <- neuralnet(revenue_binary ~ PageValues + BounceRates+Weekend+ExitRates, data = train.df, linear.output = F, hidden = 3)

plot(nn, rep="best")

nn.pred <- predict(nn, valid.df, type = "response")

3.5.1. Diagrams



```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0 3951  342
         1  198  441

               Accuracy : 0.8905
                 95% CI : (0.8815, 0.8991)
    No Information Rate : 0.8412
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5571

 Mcnemar's Test P-Value : 7.568e-10

            Sensitivity : 0.9523
            Specificity : 0.5632
         Pos Pred Value : 0.9203
         Neg Pred Value : 0.6901
             Prevalence : 0.8412
         Detection Rate : 0.8011
   Detection Prevalence : 0.8704
      Balanced Accuracy : 0.7577

       'Positive' Class : 0
```

3.5.2. Findings

The classification of customers into revenue generating and non- revenue generating can prove beneficial to evaluate a new customer. Special online promotional codes, marketing ads can be displayed to such customers, which will end up generating revenue for the company and will also cut costs by targeting revenue generating customers only.

4. Conclusions

- For our Classification model we observed that we Got An accuracy rate of about 90%,
- We were able to predict that the consumer was more likely to make a purchase based on our input data
- We also can increase the chance of sales by focusing on Three main metrics; higher page value and lower bounce rate and Paying attention to the types of products listed on pages of Administrative Type.
- We can target the predicted revenue generating customers using promotional offers and marketing ads.

5. References

1. Sakar, C.O., Polat, S.O., Katircioglu, M. et al. Neural Comput & Applic (2018). [Web Link]
2. Data Set- UCI Machine Learning Repository: Online Shoppers Purchasing Intention Dataset Data Set